

Imad EL BADISY

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PROFILE

Research Engineer in Biostatistics and Machine Learning, PhD, working at the interface of computational biostatistics and machine learning. Moroccan and French national, husband and father. I hold a PhD from Aix-Marseille University (SESSTIM laboratory) on missing data and machine learning imputation methods for survival analysis. My work focuses on building methodological tools for health data science supporting biomedical research projects. I develop open-source R packages for survival analysis, missing data, and interpretable machine learning, and contribute as a consultant and trainer in biostatistics and health data science.

EXPERIENCE

Mohammed VI Center for Research and Innovation (CM6RI)

Rabat, Morocco

Head of AI and Data Science Service

Nov. 2021 – Present

- Leading the AI and Data Science service
- Conducting methodological and applied research in health (oncology, fertility, other biomedical domains)

The GRAPH Network

Switzerland / Remote

Data Science Education Officer

Sep. 2023 – Dec. 2023

- Designed and delivered training programs in data science for global health
- Developed educational material and course content for data science learners

Institut Mines-Télécom, IMT Atlantique

Brest, France

Research Engineer in Applied Statistics

Sep. 2019 – Sep. 2021

- Conducted statistical analyses of national health data (SNDS-based analyses)
- Implemented automated and reproducible data workflows

INSERM

Clermont-Ferrand, France

Biostatistician

Jan. 2019 – Jun. 2019

- Conducted cost-effectiveness evaluation of a clinical protocol on ketamine for chronic pain

Université de Sherbrooke

Sherbrooke, Quebec, Canada

Health Economist Intern

Feb. 2017 – Sep. 2017

- Conducted a systematic review and meta-analysis on health utilities in cancer patients

EDUCATION

Aix-Marseille University, SESSTIM

Marseille, France

PhD in Biostatistics

2021 – 2025

Thesis: Missing Data and Machine Learning Methods for Survival Analysis · Supervisor: Pr Roch Giorgi

Aix-Marseille University, SESSTIM

Marseille, France

Master 2 in Quantitative Methods for Health Research

2018 – 2019

Université Clermont Auvergne

Clermont-Ferrand, France

Master 1 in Applied Mathematics and Statistics

2017 – 2018

Université Clermont Auvergne, CERDI

Clermont-Ferrand, France

Master 2 in Health Economics

2017

SOFTWARE (R PACKAGES)

[survalis](#) – Interpretable Survival Machine Learning Framework

CRAN · v0.7.1

- 19 Survival learners, 7 evaluation metrics, 8 interpretability methods · Maintainer

funcml — Functional Machine Learning Framework	CRAN
• Unified framework for functional machine learning in R · Maintainer	
survdnn — Deep Neural Networks for Survival Analysis (R torch)	CRAN
• Cox, AFT, and discrete-time neural survival models · Maintainer	
tvrmst — Time-Varying RMST from Survival Matrices	CRAN
• Dynamic RMST summaries for individualized survival curves · Maintainer	
unsurv — Unsupervised Clustering of Individualized Survival Curves	CRAN
• Phenotyping patients from predicted survival curves · Maintainer	
mcstatsim — Monte Carlo Statistical Simulation Tools	CRAN
• Functional approach to monte carlo simulation studies in R · Maintainer	
missCforest — Ensemble Conditional Trees for Missing Data Imputation	CRAN
• Missing data imputation using conditional forest ensembles · Maintainer	

PROJECTS

ML4Health — Machine Learning for Health Training	2026
• Training program focused on applied machine learning for health data analysis	
AMRPRED — Antimicrobial Resistance Prediction with Machine Learning	2026
• Predictive modeling of antimicrobial resistance using clinical and microbiological data	
MELPHY — Melanoma Phenotyping under Immunotherapy and Systemic Treatment	2025 – 2026
• Data-driven phenotyping of melanoma patients treated with immunotherapy and systemic therapies	
AIDEM — Artificial Intelligence for Dementia Prognosis and Risk Stratification	2026 – 2027
• AI-based prognostic modeling and risk stratification in dementia using curated clinical data	
CLAVUS — Predicting the Clinical Resolution of Variants of Uncertain Significance Using Machine Learning	2026 – 2027
• Machine learning approach for predicting clinical resolution of variants of uncertain significance using curated genomic data	
SURVDNN — Survival Deep Neural Networks	2025 – 2026
• Deep learning framework for survival analysis using neural networks	
OMOUMATI — Risk Prediction of Pregnancy Complications	2025 – 2026
• Prediction of pregnancy complications to support maternal and infant health outcomes	

SELECTED PUBLICATIONS

- El Badisy I.**, Assarag B., Belrhiti Z. (2026). Interpretable clinical decision support in high-risk pregnancy: A scoping review. *BMC Pregnancy and Childbirth*.
- Kadi C., Ahmadi N., Houdou A., **El Badisy I.**, et al. (2025). Differentiating latent from active tuberculosis through activation phenotypes and chemokine markers. *Biomarker Insights*, 20, 11772719241312776.
- El Badisy I.**, Graffeo N., Khalis M., Giorgi R. (2024). Multi-metric comparison of ML imputation methods: application to breast cancer survival. *BMC Medical Research Methodology*, 24(1):191.
- El Badisy I.**, BenBrahim Z., Khalis M., et al. (2024). Risk factors for colorectal cancer survival in Morocco: interpretable ML approach. *Scientific Reports*, 14(1):3556.
- Houdou A., **El Badisy I.**, Khomsi K., et al. (2024). Interpretable ML for forecasting air pollution: a systematic review. *Aerosol and Air Quality Research*, 24(1):230151.
- Zbiri S., Belghiti Alaoui A., **El Badisy I.**, et al. (2024). Private hospitals in LMICs: a typology using cluster methods — Morocco. *BMC Health Services Research*, 24(1):1231.
- Blom I.M., Eissa M., Mattijsen J.C., **El Badisy I.**, et al. (2023). Greenhouse gas mitigation interventions for health-care systems. *Bulletin of the World Health Organization*, 102(3):159.

Khalis M., Toure A.B., **El Badisy I.**, et al. (2022). Meteorological parameters and COVID-19 in Casablanca, Morocco. *Int. J. Environmental Research and Public Health*, 19(9):4989.

TEACHING

AI for Public Health — Aix-Marseille University, SESSTIM 2022 – Present

- Machine learning methods and NLP for public health · Master's students

Introduction to Biostatistics — University Mohammed VI of Health Sciences (UM6SS) 2021 – Present

- Descriptive and inferential statistics, statistical reasoning for health sciences · 100+ hours/year

Quantitative Epidemiology and Machine Learning — Institut Mines-Télécom, IMT Atlantique 2020 – Present

- ML fundamentals, quantitative epidemiology, supervised and unsupervised learning · 21+ hours/year

Outside working hours: {**Methods in Health Data Science with R**} — an independent online course pathway covering applied biostatistics, survival analysis, machine learning, and meta-analysis with R.

SELECTED TALKS

Comparison of missing data imputation methods in Cox regression with time-varying effects 2023
ISCB 2023, Rome · El Badisy I. et al.

Missing data imputation using a meta-algorithm metaCART: simulation under MCAR 2023
EpiClin 2023, Nancy · El Badisy I. et al.

SKILLS

Methods: Biostatistics · Survival analysis · Missing data · Meta-analysis · Interpretable machine learning

Tools: R package development · Python · Quarto/LaTeX · Git/GitHub · Slurm/HPC

Languages: Arabic (native) · French (fluent) · English (fluent)